

Supplementary Information

Ayoub Lasri

Berache Limited, Dublin, Ireland

Mechanistic simulation of heritable expression states and drug tolerance in growing, dividing single-cell populations. Supplementary Notes 1–14, Supplementary Figures 1–6 and Supplementary Tables 1–5.

Supplementary Note 1: Stationary laws used for validation

Birth-death. For $\emptyset \xrightarrow{k} X \xrightarrow{\gamma} \emptyset$ the stationary distribution is Poisson with mean k/γ .

Telegraph model. With activation k_{on} , inactivation k_{off} , transcription k_{tx} from the active state and degradation γ , and $a = k_{\text{on}}/\gamma$, $b = k_{\text{off}}/\gamma$, $\lambda = k_{\text{tx}}/\gamma$, the stationary distribution is the Beta-Poisson mixture

$$P(n) = \frac{\lambda^n \Gamma(a+n) \Gamma(a+b)}{n! \Gamma(a+b+n) \Gamma(a)} {}_1F_1(a+n; a+b+n; -\lambda),$$

whose mean is $\lambda a/(a+b)$ and whose variance is $\mu + \lambda^2 ab/[(a+b)^2(a+b+1)]$. We evaluate ${}_1F_1(\alpha; \beta; -\lambda)$ through Kummer’s transformation $e^{-\lambda} {}_1F_1(\beta - \alpha; \beta; \lambda)$, whose series has positive terms; partial sums are rescaled by 10^{-200} whenever they exceed 10^{200} so that the log of the sum is accumulated without overflow.

Bursty protein. Protein produced in geometric bursts of mean size b arriving at rate a and degraded at rate γ is negative binomial with shape a/γ and success probability $1/(1+b)$. The package emulates bursts through an mRNA of lifetime $1/(200\gamma)$ translated at rate $200\gamma b$.

Supplementary Note 2: Kernel details and correctness tests

The direct method draws the waiting time from $\text{Exp}(a_0)$ and the channel by linear search over the cumulative propensities; after each event only the propensities of reactions that depend on a changed species (the dependency graph) are recomputed, and the running total is refreshed every 512 events. Fixed-step tau-leaping draws $\text{Poisson}(a_j\tau)$ firings for every channel; the strict variant errors when a species would become negative, the hybrid variant discards the leap and simulates that interval with the direct method. Discarding is conditioned on the realised leap, so the accepted steps of the hybrid variant follow a truncated Poisson law and the kernel is approximate; Fig. 2d measures the resulting discrepancy against the exact stationary laws at two step sizes. Adaptive tau-leaping classifies as critical every channel within $n_c = 10$ firings of exhausting a reactant, selects τ from the non-critical channels through the bounds $\max(\varepsilon x_i/g_i, 1)/|\mu_i|$ and $\max(\varepsilon x_i/g_i, 1)^2/\sigma_i^2$, draws the time to the next critical event from $\text{Exp}(a_0^{\text{crit}})$, fires at most one critical channel, halves τ on rejection, and performs 100 exact events whenever $\tau < 10/a_0$.

Supplementary Table 1 lists the correctness tests of the package test suite. They run on every commit under continuous integration, so the tolerances below are enforced rather than reported once.

Supplementary Table 1. Correctness tests of the kernels and of the population layer (`test/test_kernels.jl`,
`test/test_population.jl`, `test/test_exact_population.jl`).

Model	Kernel	Cells	Statistic	Tolerance
Birth-death, $k = 12$, $\gamma = 1$	direct	8,000	Kolmogorov- Smirnov distance to Poisson(12)	< 0.015
same	hybrid ($\tau = 0.02$), adaptive, strict tau-leap ($\tau = 0.005$)	8,000 each	same	< 0.02
same	all four	8,000 each	error of the mean; minimum count	< 0.2 ; ≥ 0
Telegraph, $k_{\text{on}} = 0.4$, $k_{\text{off}} = 0.6$, $k_{\text{tx}} = 12$	direct	12,000	Kolmogorov- Smirnov distance to the Beta-Poisson law	< 0.015
same	direct	12,000	error of the mean transcript number; of the active-promoter fraction	< 0.15 ; < 0.02
Bursty protein, $a = 1.5$, $b = 6$	direct	8,000	Kolmogorov- Smirnov distance to NegBin(a/γ , $1/(1 + b)$)	< 0.03
Stiff dimerisation ($k = 50$, $k_d = 0.05$, $k_u = 1$, $g = 0.5$)	adaptive ($\varepsilon = 0.02$) against direct	2,000 each	relative difference of the mean dimer number	$< 5\%$
Birth-death	strict tau-leap ($\tau = 0.5$) on a model that would go negative	400	an error is raised rather than a negative count returned	raised
Birth-death	direct, one thread against several	300	identical final states	exact
Telegraph in a dividing population	hybrid	200	population size held constant under ConstantN ; doubling time under FreeGrowth	exact; within 10%
Constitutive production with a memoryless timer	direct	3,000 lineages	mean and standard deviation against the closed-form single-lineage law	< 0.5 each

Model	Kernel	Cells	Statistic	Tolerance
same	direct	branching population	mean against the closed-form population law; separation from the lineage law	< 1.0 ; > 1.5
Gamma interdivision times, $cv = 0.25$ and 1	—	20,000 draws	error of the mean and of the coefficient of variation	< 0.15 ; < 0.05

32 Supplementary Note 3: Parameters of all simulations

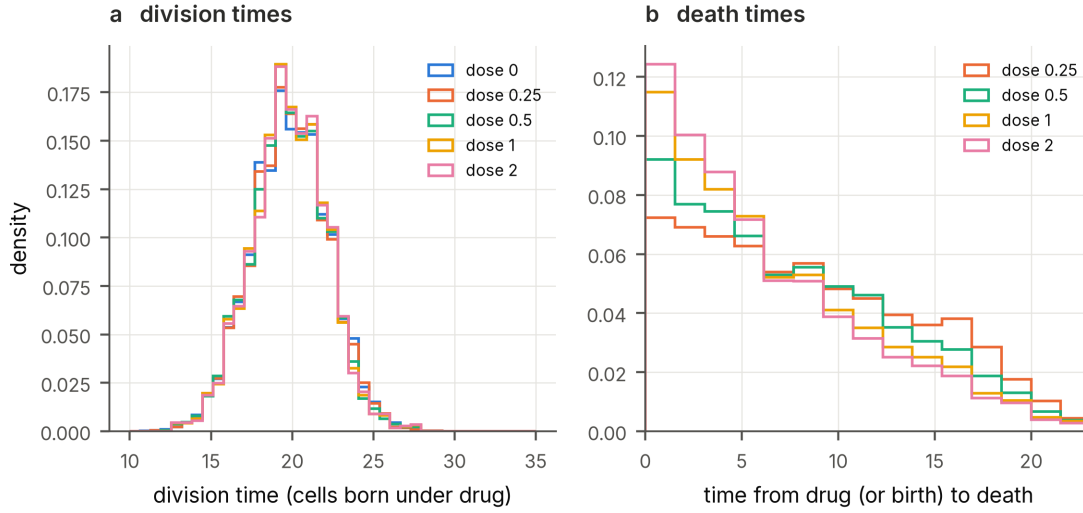
33 Supplementary Table 2. Parameters of every simulation in this Article.

Figure	Model	Parameters
2a	birth-death	$k = 12, \gamma = 1, T = 15$
2b	telegraph	$(k_{\text{on}}, k_{\text{off}}, k_{\text{tx}}) = (0.4, 0.6, 12)$ and $(0.05, 0.5, 40), \gamma = 1, T = 40$
2c	bursty protein	$a = 1.5, b = 6, \gamma = 1, T = 25$
2d-e	telegraph	$(0.4, 0.6, 12), n = 20\,000$
2f	random telegraph GRNs	$n_{\text{act}} = n_{\text{genes}}, n_{\text{inh}} = n_{\text{genes}}/2,$ $\lambda = \ln 2/20, dt = 0.1, T = 20$
3	telegraph gene with protein	$k_{\text{tx}} = 30, k_{\text{dm}} = 1, k_{\text{tl}} = 4,$ $k_{\text{dp}} = 0.2, k_{\text{on}} = k_{\text{off}}$ as indicated; $\lambda = \ln 2/20, \text{size } V_{\text{div}} = 2$ (cv 0.05), $\sigma_f = 0.02$, replication at 50% of the cycle where indicated; panel 3d varies the cell-cycle time over 10, 20 and 40; panel 3f uses $(k_{\text{on}}, k_{\text{off}}, k_{\text{tx}}) = (0.5, 1.5, 40)$ with 3,000 cells
3e	stable protein	$k_{\text{on}} = k_{\text{off}} = 5, k_{\text{tx}} = 20, k_{\text{dm}} = 1,$ $k_{\text{tl}} = 2, k_{\text{dp}} = 0.05$

Figure	Model	Parameters
4	resistance gene	as Fig. 3 with $(k_{\text{on}}, k_{\text{off}}) = (0.002, 0.02)$ (memory, $p_{\text{on}} \approx 9\%$) or $(0.05, 0.5)$ (fast, same p_{on}); founders burnt in for 300 time units at constant size; death $h_{\text{max}} = 0.15$, $\text{EC}_{50} = 0.5$, $m = 2$, $K = 150$, $q = 4$; drug from $t = 40$ of the treatment stage; drug-induced variant $(0.0002, 0.02)$ with $k_{\text{on}} \rightarrow k_{\text{on}}(1 + 100d)$; fitness cost variant: growth $\times (1 - 0.5 c_P^4 / (150^4 + c_P^4))$; memory disruption: switching $\times 20$ on $[0, 40)$ or $[0, 120)$; schedule scan: 300 founders, 240 time units, 20-time-unit exposures, five independent seeds per point; cycle-gated variant (Supplementary Fig. 5) adds $\beta = 0.25$, $\varphi_0 = 0.5$, $w = 0.15$ to the death hazard, and is repeated with h_{max} raised to 0.295 so that the cycle-averaged hazard matches the cycle-blind one
5a	40 independent telegraph genes	$k_{\text{on}} = k_{\text{off}} \in [10^{-3}, 1]$ (log-spaced), $k_{\text{tx}} = 20$, $k_{\text{dm}} = 0.5$; 40 founders, 6 doublings
5b-c	random GRN	30 genes, 30 activations, 15 inhibitions, $k_{\text{tx}} \in [5, 30]$, $K \in [2, 10]$, $n = 2$, basal 0.05; 1,500 cells; sequencing capture 0.15 (cv 0.3)
5d	random GRN	20 genes, 22 activations, 10 inhibitions; knockdown factor 0.05; 800 cells
6a-b	telegraph	truth $(0.3, 0.6, 20)$; priors log-uniform on $[0.01, 10]$, $[0.01, 10]$, $[1, 200]$; 200 (a) / 150 (b) particles
6c	resistance gene	truth $h_{\text{max}} = 0.5$, $K = 150$; priors log-uniform on $[0.05, 5]$ and $[20, 1000]$; 100 particles

Figure	Model	Parameters
7	resistance gene, hours	cell cycle 24 h (sizer, cv 0.1), $k_{tx} = 30$, $k_{dm} = 1$, $k_{tl} = 4$, $k_{dp} = 0.2$ per h, $K = 150$, $q = 4$, growth-arrest Hill coefficient $m_g = 2$; fitted: k_{on}, $k_{off} \in [10^{-3}, 0.3]$, $h_{max} \in [0.005, 0.5]$ per h, $EC_{50} \in [3, 40]$ μM, death Hill coefficient $m \in [1, 6]$, $IC_{50} \in [1, 40]$ μM; 300 founders burnt in for 10 cycles, 48 h drug-free, 72 h drug, integration step $dt = 0.5$ h; 160 Latin-hypercube points and 80 Nelder-Mead iterations with seed 1; identifiability (Supplementary Fig. 3) by 11-point log-grid conditional profiles of the six parameters, a 7×7 memory $\times$ resistant-fraction slice, and the step halved to 0.25 h with three seeds, via the profile part of the script; the cycle-dependent variant (Supplementary Fig. 4) adds a seventh parameter $\beta \in [0, 1]$ with $\varphi_0 = 0.5$, $w = 0.15$, fitted under the same budget by the cycle part
S6	exact-solution validation	constitutive $k = 20$, $\gamma = 1$; bursts of $b = 4$ at k/b; telegraph $(k_{on}, k_{off}, k_{tx}) = (0.5, 1, 40)$, all with a memoryless interdivision time of mean 1, $dt = 0.002$, exact kernel, 40,000 lineages and five populations capped at 15,000 cells, burnt in for 12 generations; replication model $k = 20$, $\gamma = 1$, volume-scaled synthesis, exponential growth $\ln 2$, deterministic cycle of 1, replication at 50% of the cycle, founders placed uniformly on the update grid, 14 generations; step-size scan over $dt = 0.02, 0.01, 0.005, 0.002$

34 Supplementary Note 4: Additional persister simulations



35

36 **Supplementary Fig. 1 | Single-cell timing distributions underlying Fig. 4c.** **a**, Division times of cells
 37 born under drug and **b**, times from drug start (or from birth, for cells born under drug) to death, per dose.

38 The division-time distribution is set by the sizer and is the same at every dose, because in this model the drug
 39 kills but does not slow growth. Times to death are broad (coefficient of variation 0.72 at dose 0.5 and 0.82 at
 40 dose 2) and their mean shifts by much less than the population decay rate, because most deaths occur among
 41 low-expressing cells whose hazard is already close to $h_{\max}$ at dose 0.5 ($EC_{50} = 0.5$, $m = 2$); raising the dose
 42 mainly shortens the survival of cells with intermediate protection. The dose dependence of population decay is
 43 therefore carried by the fraction of cells that are protected, not by the kinetics of death of unprotected cells,
 44 which is the interpretation [1] give of their single-cell tracking data.

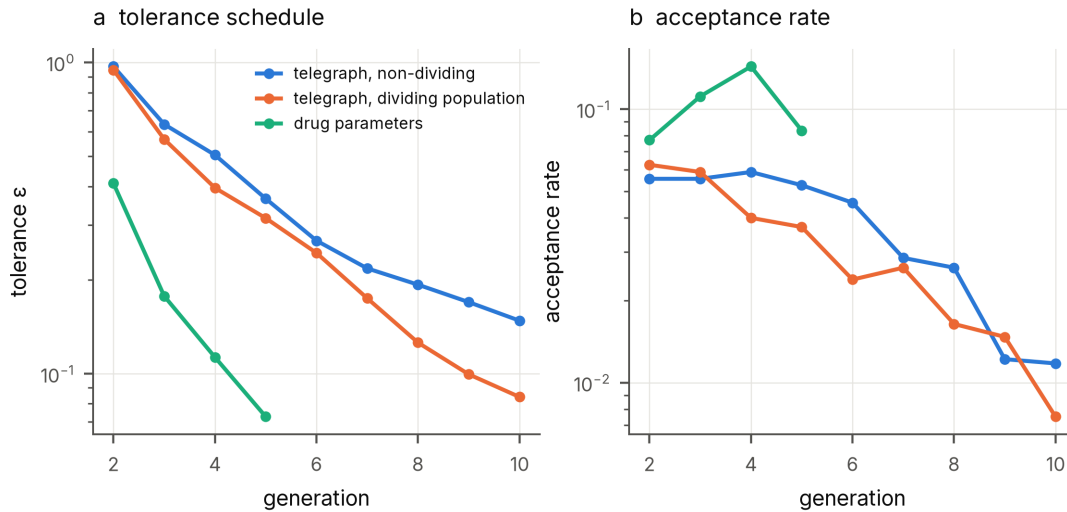
45 Under continuous dosing at the highest dose (release period 0, dose 2 in Fig. 4h), the long-term growth rate is
 46 +0.0038 per time unit for pre-existing tolerance, -0.0151 when the resistant state carries a 50% growth cost and
 47 +0.0123 for drug-induced tolerance; the schedules with the lowest long-term growth rate are release period 0,
 48 dose 2 (growth rate +0.0038 per time unit, ± 0.0007 over 5 seeds), release period 0, dose 2 (growth rate -0.0151
 49 per time unit, ± 0.0012 over 5 seeds) and release period 0, dose 1 (growth rate +0.0089 per time unit, ± 0.0001
 50 over 5 seeds) respectively. For pre-existing tolerance without a cost, every release period raises the net growth
 51 rate; with the fitness cost, release periods of 5 and 10 time units are within 0.003 per time unit of continuous
 52 dosing at doses of 1 and 2; for drug-induced tolerance, dose 1 gives a lower net growth rate than dose 2 at every
 53 release period, and release periods of 20 time units or more raise the growth rate in every model. Each point is a
 54 population of 300 founder cells. Each point of the scan is the mean of 5 independent seeds, and every schedule
 55 is run on the same 5 seeds, so schedules are compared by their paired differences, in which the shared founder
 56 population cancels; the spread quoted beside each winner above is the seed-to-seed scatter of its absolute growth
 57 rate, which is 0.5 to 2.0 times the paired differences and is not the error on them. The winning schedule beats the
 58 best schedule with a non-zero release period by 0.00026 to 0.00142 per time unit against paired standard errors of
 59 0.00013 to 0.00028, a margin that a two-sided paired t-test resolves at the 5% level in 1 of the 3 mechanisms, the
 60 one in which the resistant state carries a fitness cost. Elsewhere the scan identifies a region of good schedules
 61 rather than a single best one (pre-existing tolerance +0.00037 against 0.00023, $t = 1.61$ on 4 d.f., $P = 0.18$;
 62 drug-induced tolerance +0.00026 against 0.00013, $t = 2.02$ on 4 d.f., $P = 0.11$).

63 Supplementary Note 5: Benchmark details

64 Network inference methods: absolute Pearson and Spearman correlations of $\log(1+x)$ (Pearson) or ranks
 65 (Spearman); GENIE3 as random-forest importances (200 trees, `sqr`t features) fitted on $\log(1+x)$ with scikit-learn.
 66 Imputation: kNN-smoothing (one step, $k = 15$, 10 principal components of the Freeman-Tukey transformed, library-
 67 size normalised counts) and MAGIC with default parameters on square-root normalised counts. Metrics: area
 68 under the precision-recall and receiver operating characteristic (ROC) curves against the undirected (correlation)
 69 or directed (GENIE3) ground-truth adjacency. The cell-cycle-regressed dataset is the residual of $\log(1+c)$
 70 (concentration c) after ordinary least-squares regression on gene copy number, cell age and age squared, per gene.

71 Supplementary Note 6: Inference diagnostics

72 **Summary statistics and distances.** For Fig. 6a,b the distance between a simulated and the observed sample
 73 of counts is their Wasserstein-1 distance [2], computed as the mean absolute difference between the two empirical
 74 quantile functions on a grid of 201 probabilities, divided by the observed mean; each simulated sample contains
 75 600 cells (a) or the final snapshot of 500 founders grown for 80 time units (b), against 2,000 observed cells. The
 76 distance between two independent samples at the true parameters (the noise floor) is reported in the summary
 77 tables of the repository. For Fig. 6c the summaries of a treated population are the log surviving fraction on a
 78 grid of 25 time points from the start of the drug to 60 time units later (populations that went extinct contribute
 79 a floor of 10^{-3}) and the concordance of death fates between sisters born in the 20 time units before the drug; the
 80 distance is the root-mean-square difference of these summaries. Each generation's tolerance is the median of the
 81 previous generation's accepted distances, particles are perturbed with a Gaussian kernel of variance twice the
 82 weighted variance of the previous generation, and the first generation samples the prior.



84 **Supplementary Fig. 2 | ABC-SMC diagnostics.** a, Tolerance and b, acceptance rate per generation for the
 85 three runs of Fig. 6.

86 The run on non-dividing cells (10 generations, 200 particles) reached a final tolerance of 0.148 with an acceptance
 87 rate of 1% in the last generation; the division-aware run (10 generations, 150 particles, each particle simulating a
 88 population of 500 founders for 80 time units) reached 0.0841 at 1%; the drug-parameter run (5 generations, 100
 89 particles, 200 founders each) reached 0.0727 at 8%. The importance weights of the final generation are uneven,
 90 so fewer particles carry the posterior than were run: by Kish's formula the effective sample sizes of the final

generations are 59 of 200 for the non-dividing run, 19 of 150 for the division-aware run and 75 of 100 for the drug-parameter run. The posteriors of Fig. 6a,b are therefore drawn with a few tens of effectively independent particles, and the isolated tall bars separated by empty bins in those histograms are the weight of one or two particles rather than structure in the posterior. They are not a boundary effect: the accepted values run over k_{on} 0.18 to 0.885, k_{off} 0.219 to 9.47, k_{tx} 14.1 to 86.7 in the non-dividing run and k_{on} 0.181 to 0.793, k_{off} 0.239 to 9.34, k_{tx} 15.4 to 74.8 in the division-aware run, against prior supports of 0.01 to 10 on both switching rates and 1 to 200 on the transcription rate, so no particle is clipped at a prior edge and the histograms of Fig. 6a,b are binned over the whole prior support. In the drug-parameter run the accepted values are h_{max} 0.241 to 4.96, K 70.7 to 181, and the upper end of h_{max} does approach the prior bound of 5, because the ridge of Fig. 6c runs to the edge of the prior. Posterior predictive checks were made by simulating the forward model at the posterior median (Fig. 6d for the drug parameters); the posterior median rather than the mean is reported throughout because the posteriors are asymmetric on the log scale.

Supplementary Note 7: Runtime

Supplementary Table 3 lists the wall-clock times of the population simulations of Fig. 2f (200 steps of $dt = 0.1$ for random telegraph networks of 10, 50 and 100 genes, four threads of an Intel Xeon processor at 2.1 GHz, Julia 1.10). Throughput in cell-steps per second falls with the number of reactions because every step of every cell has to sample the events of all channels; the hybrid kernel is 2 to 5 times faster than the direct method at these tolerances, the gap widening with the number of reactions. The single-cell interface used for the comparison with JumpProcesses.jl (Fig. 2e) allocates a workspace per simulation and is slower in serial execution than JumpProcesses.jl on that one-gene model; the population interface reuses one workspace per thread.

Supplementary Table 3. Runtime of population simulations.

genes	cells	kernel	threads	seconds (200 steps)	cell-steps / s
10	100	DirectSSA	4	0.38	52,398
10	100	HybridSSATau	4	0.11	180,019
10	1000	DirectSSA	4	1.04	193,080
10	1000	HybridSSATau	4	0.58	346,004
10	5000	DirectSSA	4	4.78	209,388
10	5000	HybridSSATau	4	2.57	388,683
50	1000	DirectSSA	4	9.72	20,581
50	1000	HybridSSATau	4	2.57	77,671
100	1000	DirectSSA	4	31.41	6,367
100	1000	HybridSSATau	4	6.37	31,373

Supplementary Note 8: Laboratory and clinical reference data

Supplementary Table 4. Reference values used for calibration and validation (transcribed from the cited articles; files in `paper/data/`).

Quantity	Value	Source
U2OS cells tracked at cisplatin addition (7 / 10 / 13 μ M)	232 / 240 / 296	[1], Table 2
of which died within 72 h	30 / 66 / 176	same
of which divided within 72 h	100 / 75 / 67	same
of which survived without dividing	102 / 99 / 53	same
HCT116 cells at drug addition; death fraction	275; 0.64	[1], Table 1
Plateau of the HCT116 kill curve	after ~100 h; the U2OS curves show no biphasic decay at any of the three concentrations	[1], Fig 1b,e
Lineage correlations of fate (HCT116 only; the U2OS dataset tracked one randomly chosen daughter per division, so sister and cousin pairs cannot be formed in it)	present for first and second cousins; absent for third cousins (reported graphically; no numeric coefficients given); the memory of at least two to three generations these imply sets the prior used in the calibration	[1], Fig 5a
Single-cell intermitotic and apoptosis times across doses (U2OS)	not significantly different (Kruskal-Wallis $P = 0.22$ and $P = 0.53$) while population decay rates differ about threefold	[1]
Pre-resistant melanoma cells traced back from resistant fates	initial frequency ~1:1,000 to 1:10,000 (context, not a value used: the simulations here use 1:200; Methods)	[3]
Pre-resistant melanoma cells	1:50 to 1:500 per marker; EGFR-high cells give 7.9 ± 0.9 fold more resistant colonies	[4]

Supplementary Note 9: Migration from version 1

The v1 API (`Donne`, `ssa`, `tauleap`, `tauleapswitch`, `adaptive_tauleap`, `exponential_growth`, `NamedTuple` reactions) remains available through a deprecated compatibility layer that converts `NamedTuple` reactions to `ReactionModels` (reactions whose names contain `act`, `inhib`, `comb_a` or `comb_i` become Hill kinetics with `rate = [k, n, K]`; species whose names contain `on/off` become promoter groups) and maps the algorithms onto the new kernels. Output shapes are preserved, including the leading `:NULL` column. See `docs/src/migration.md`.

Supplementary Note 10: Relation to previous work

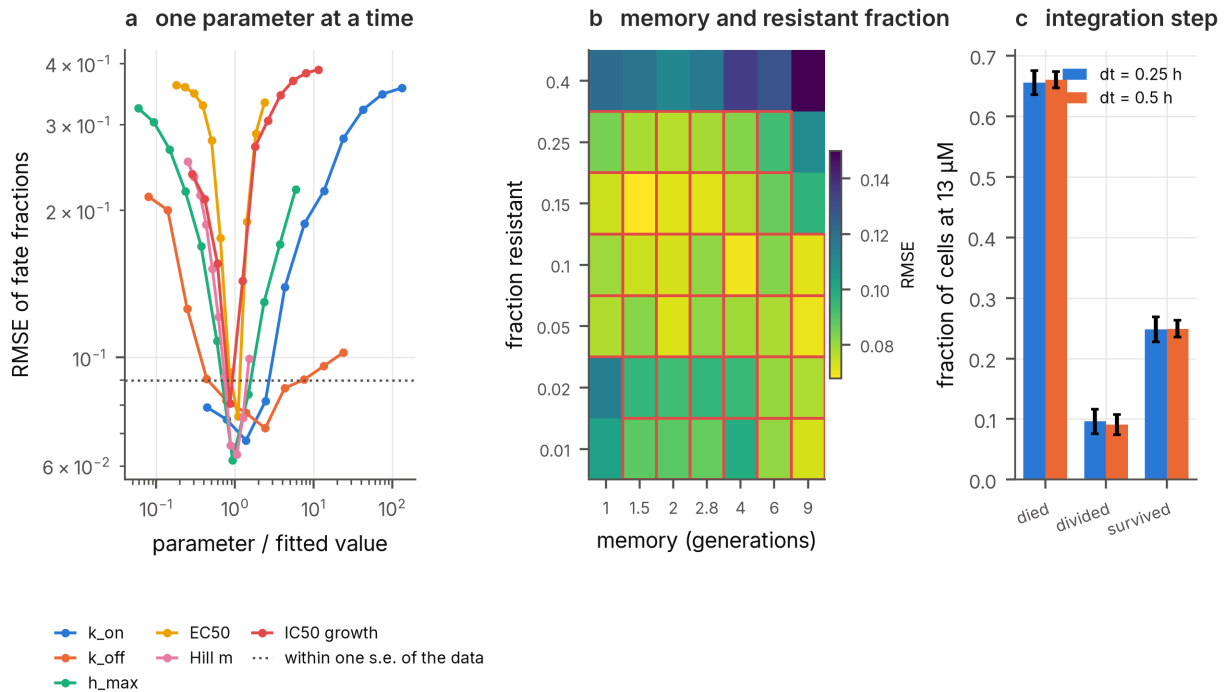
The ingredients of this framework have precedents, and several of its results confirm conclusions reached earlier by other means. The table separates, claim by claim, what was already established from what this work adds. Prior results the framework reproduces are evidence that it is behaving correctly, not claims of novelty.

Result in this Article	Already established	What this work adds
Stochastic reaction kinetics inside growing, dividing cells (Fig. 3)	Simulators of stochastic expression with growth and division exist [5–7], including version 1 of this framework [8]	Gene replication, promoter inheritance, a drug layer, lineage statistics, observation models and inference in one forward model, so that a resistance phenotype is a heritable expression state with measurable kinetics rather than an assumed compartment
Synthetic scRNA-seq from a known network in a dividing population, for imputation and network-inference benchmarking (Fig. 5)	Version 1 of this framework already did this [8]	Gene replication and cell-cycle copy number, sizer and adder division rules in place of a timer, inherited promoter states, the lineage table, the drug layer, likelihood-free inference and the schedule optimiser; nothing in Fig. 5 that version 1 could also produce is presented as new
Exact stationary distributions for growing, dividing cells differ between a single lineage and a population snapshot (Supplementary Note 14)	Solved exactly for bursty expression with replication, partitioning and general interdivision times [9], for the extended telegraph model with volume-dependent synthesis and size control [10], and framed as an ergodic principle [7,11]	The population and lineage layers are validated against those solutions in both modes, rather than only against single-cell stationary laws
Inference of kinetic parameters from lineage-resolved data	Finite-state-projection inference from mother machines and lineage trees [12]	Inference of drug parameters from population data, with the identifiability limit made explicit (Fig. 6c)
Simulation of division trees for lineage-tracing methods	TedSim couples expression to division history [13]; Cassiopeia simulates topologies, heritable fitness and CRISPR barcodes [14]	Molecular content on the same tree: partitioning at division, promoter states, drug-induced death, and per-cell fates with the molecules that caused them
Heritable expression states decide which cells survive a drug (Fig. 4)	Measured directly in barcoded and time-lapse experiments [1,4,15,16]	A generative model in which memory is one promoter timescale, reproducing the reported signatures quantitatively and predicting which of them discriminate pre-existing from induced tolerance
Cell-cycle-aware inference of transcriptional kinetics is necessary (Fig. 3f, Fig. 6b)	Established from data and theory [17–19]	A simulator that generates the data such methods assume, and a quantification of the bias incurred by a division-blind fit

Result in this Article	Already established	What this work adds
Benchmarks of network inference from single-cell data	Extensive benchmarks exist [8,20,21]	The growth and cell-cycle confound with known ground truth: how much accuracy is lost in a dividing population, and how much cell-cycle regression recovers (Fig. 5b)
Perturbing the retention of a resistant state changes the outcome (Fig. 4f)	Phenotypic-switching theory [22] and memory-disrupting compounds [15]	The timing requirement (disruption must continue during exposure) and a lineage-resolved readout, the number of surviving clones
An intermediate dose can be optimal against drug-induced tolerance (Fig. 4h)	Reported for induced persisters [23]	Reproduced; the accompanying benefit of release periods does not appear here, which the text attributes to the decay of the induced state in this model rather than to a disagreement about data

Supplementary Note 11: Identifiability and numerical robustness of the calibration

Six parameters were fitted to six measured fractions, so the fit cannot determine all six. Three checks bound what the calibration does and does not establish.



Supplementary Fig. 3 | What the fate fractions determine. a, Root-mean-square error of the simulated fate fractions at the training concentrations when one parameter is moved away from its fitted value over its

132 full range and the others are held at theirs, plotted against the ratio to the fitted value; the dotted line marks
133 one binomial standard error of the measurement above the best fit. **b**, The same error over the memory of the
134 resistant state and the fraction of cells in it, with the death parameters held at their fitted values; red outlines
135 mark the combinations that lie within one standard error of the best. **c**, Simulated fates at 13 μM with the
136 integration step halved (mean $\pm$ s.d. of three seeds).

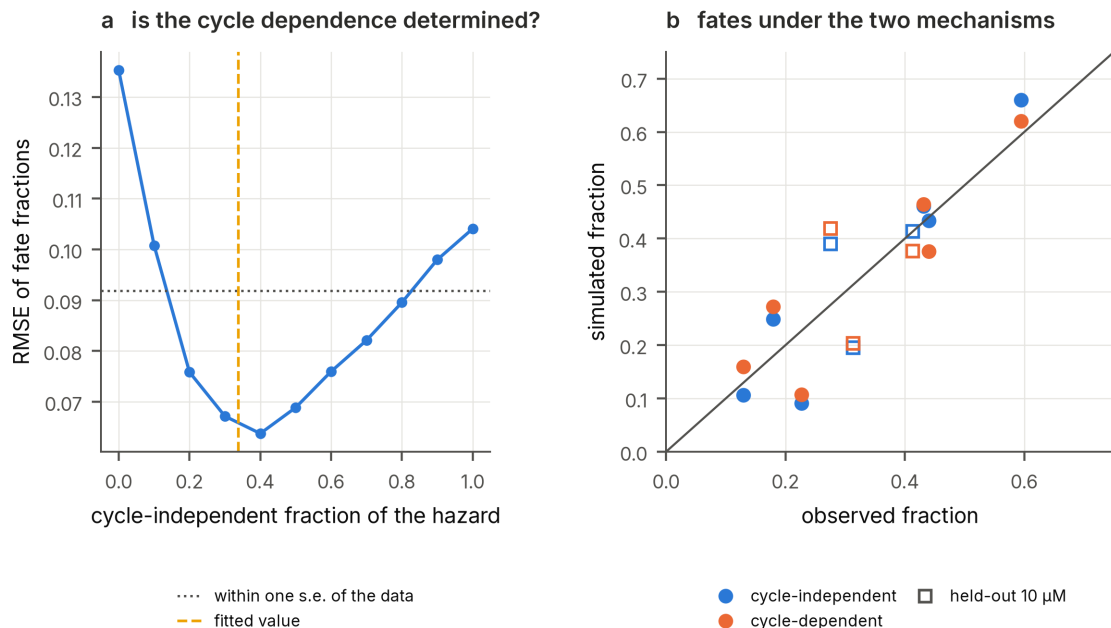
137 The death parameters are the better determined: the maximal death rate stays within one standard error of the
138 best fit over about a twofold range, and the EC_{50} , Hill coefficient and growth-arrest IC_{50} over narrower ranges
139 still (Supplementary Fig. 3a). The two promoter switching rates are not determined: each can move by roughly
140 sixfold with no penalty the data can detect. The two-dimensional slice shows why (Supplementary Fig. 3b). A
141 short memory with many resistant cells and a long memory with few produce nearly the same fate fractions, so
142 the error surface is flat along that direction, and a majority of the combinations tested (memory one to nine
143 generations, resistant fraction 1% to 40%) fit within the noise of the measurement; those that do span one to nine
144 generations of memory and 1% to 25% of cells resistant. The memory quoted in the main text is therefore set by
145 the prior taken from the lineage correlations, not by the fate fractions, and the resistant fraction that accompanies
146 it should not be read as a measurement.

147 Halving the integration step from 0.5 h to 0.25 h changes each simulated fate fraction by less than the seed-to-seed
148 spread (Supplementary Fig. 3c), so the discretisation of the death hazard is not a source of error at this step size.

149 A profile in which the remaining parameters are re-optimised at each fixed memory would be preferable to a slice,
150 but at seven seconds per objective evaluation the inner search could not be given enough budget to be reliable: it
151 returned fits three times worse than the known solution at short memory, which reflects the optimiser rather than
152 the data. The slice avoids that failure at the cost of conditioning on the fitted death parameters.

153 **Supplementary Note 12: Cell-cycle-dependent killing as an alternative** 154 **to heritable expression**

155 The lethality of cisplatin is enhanced during replication, so where a cell sits in its division cycle when the drug
156 arrives is a competing explanation for two of the observations the calibrated model reproduces: the near-invariance
157 of the mean time to death across concentrations, and the large fraction of cells that survive three days of drug
158 without dividing. A model in which killing is confined to a window of the cycle would spare exactly those cells
159 that never enter that window, without any heritable expression state.



160

161 **Supplementary Fig. 4 | Cell-cycle-dependent killing tested against the fate data.** **a**, Root-mean-square
 162 error of the fate fractions at the training concentrations as the cycle-independent fraction β of the death hazard is
 163 moved from 0 (killing confined to the replication window) to 1 (cycle-independent), with the remaining parameters
 164 held at the cycle-dependent fit; dotted line, one binomial standard error of the measurement above the best fit;
 165 dashed line, the fitted value. **b**, Simulated against observed fate fractions at all three concentrations for the
 166 cycle-independent and the cycle-dependent fit (squares, the held-out 10 μ M condition; line, equality).

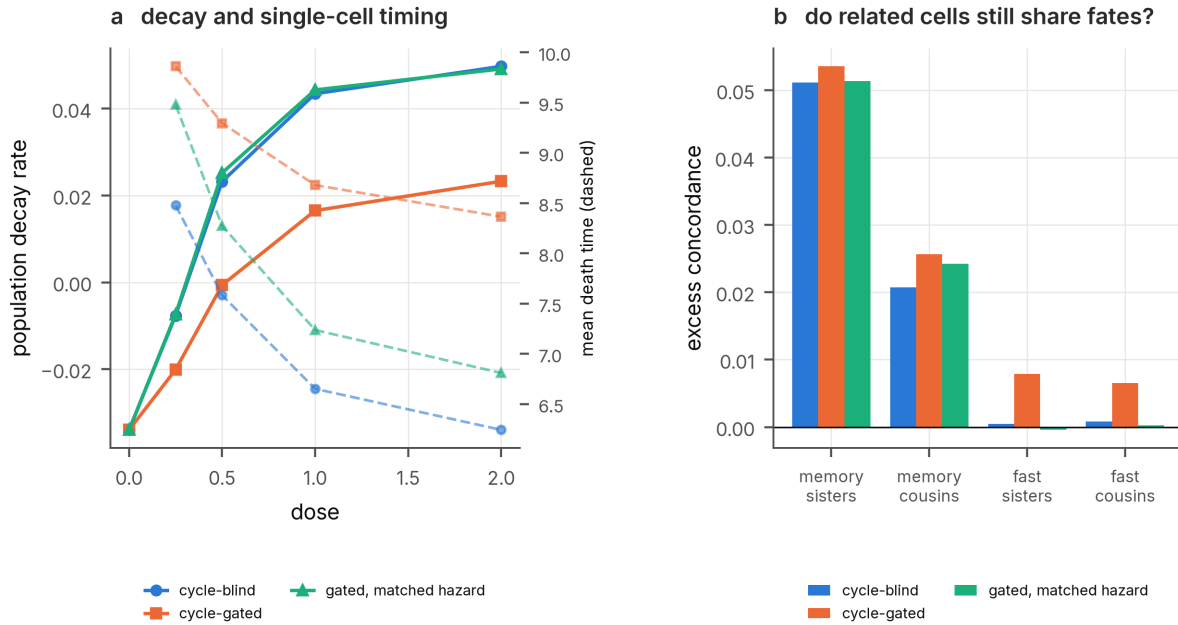
167 Refitting under the same objective, bounds and optimiser budget gives a cycle-independent fraction of 0.34 and a
 168 root-mean-square error of 0.064 on the training fractions, against 0.061 for the cycle-independent model; at the
 169 held-out concentration the errors are 0.106 and 0.095. The profile over that fraction is flat: every value from 0.20
 170 to 0.80 lies within one binomial standard error of the measurement (0.028) of the best fit (Supplementary Fig.
 171 4a). Three fate counts per concentration therefore cannot say whether cells are spared because they inherited a
 172 protective state or because they were outside the replication window, and the parameters of either fit should be
 173 read with that in mind.

174 The two mechanisms are not exclusive, and the experiment that separates them is not a fate count: it is a
 175 measurement of cell-cycle position at the moment the drug arrives, which time-lapse imaging with a cycle reporter
 176 provides directly. Until that is done, the heritable-expression reading of the calibrated parameters should be
 177 understood as one of two accounts that these fate fractions cannot tell apart.

178 **Supplementary Note 13: Cell-cycle-dependent killing in the persister** 179 **simulations**

180 Supplementary Note 12 shows that the cisplatin fate data cannot decide whether cells are spared because they
 181 inherited a protective state or because they were outside the replication window. The persister simulations of Fig.
 182 4 use a hazard with no cell-cycle dependence at all, so the same question applies to them: do their conclusions
 183 rest on that choice? We repeated them with three quarters of the death hazard confined to a window around
 184 mid-cycle ($\beta = 0.25$, $\varphi_0 = 0.5$, $w = 0.15$), leaving everything else unchanged.

185 The window position is a modelling choice rather than a claim about any particular agent: the persister study
 186 uses a generic drug and commits to no mechanism. What these runs test is therefore robustness to a strong
 187 cell-cycle gate of any kind, not the phase specificity of a named drug.



188

189 **Supplementary Fig. 5 | The persister simulations under cell-cycle-gated killing.** Three arms throughout:
 190 the cycle-blind hazard; three quarters of it gated to a mid-cycle window; and the same gate with the maximal
 191 hazard raised so that the cycle-averaged hazard matches the cycle-blind one. **a**, Net population decay rate over the
 192 first 30 time units of treatment against dose (solid, left axis) and the mean time to death of killed cells (dashed,
 193 right axis). At matched mean hazard the dose response returns to the cycle-blind curve, which is what shows
 194 that the shallower response under the gate was the lower average hazard rather than the gating. **b**, Concordance
 195 of lineage fate between sisters and between cousins, above the value expected if fates were independent, for the
 196 memory gene and the fast-switching control; at matched hazard the gate leaves no excess concordance at all in
 197 the control, which has no usable memory.

198 Population decay still rises with dose (-0.001 to +0.023 per time unit over doses 0.5 to 2, against +0.023 to
 199 +0.050 without the gate), at roughly half the rate, and the mean time to death of killed cells stays nearly
 200 dose-invariant (8.4 to 9.3 against 6.2 to 7.6 time units over the same doses), so the dose-dependence of decay
 201 alongside dose-invariant single-cell timing is not an artefact of a cycle-blind hazard. The kin correlations deserve
 202 more care, because sisters are born together and therefore occupy the same phase of the cycle: a gated hazard
 203 can in principle correlate their fates with no heritable expression state at all. It does not: in the fast-switching
 204 control, which has no usable memory, the sister concordance above independence rises only from +0.000 to +0.008,
 205 while the memory gene sits at +0.054, 7-fold larger. In the schedule scan, continuous dosing wins in all 3 models
 206 with the gate as without it, at the same dose in 3 of 3. The gate raises the long-term growth rate (+0.0026 to
 207 +0.0155 per time unit at all 60 points of the scan), which is what confining three quarters of the hazard to a
 208 window of the cycle implies, and it perturbs the order of the schedules below the winner (Spearman correlation
 209 0.977 to 0.998 between the gated and cycle-blind orderings), but it changes neither which schedule wins nor the
 210 intermediate-dose optimum under drug-induced tolerance (+0.0156 against +0.0190 under the gate and +0.0089
 211 against +0.0123 without it). Across the two scans a two-sided paired t-test resolves the margin over the best

212 holiday at the 5% level in 4 of the 6 comparisons, the exceptions being pre-existing tolerance without the gate
 213 (+0.0004 against a paired standard error of 0.0002) and drug-induced tolerance without the gate (+0.0003 against
 214 a paired standard error of 0.0001).

215 The gate does two things at once. It makes the hazard depend on cycle phase, and it lowers the hazard on average:
 216 over a uniform cycle phase the multiplier averages 0.532, and under the phase density these simulations realise,
 217 $f(\varphi) = 2/(1 + \varphi)^2$ for volume-based cycle progress under a sizer with exponential growth in a growing population,
 218 0.508. Half of the shallower dose response above is therefore just less killing. Repeating the comparison with the
 219 maximal hazard raised by $1/0.508$, so that the cycle-averaged hazard matches the cycle-blind one ($h_{\max} = 0.295$
 220 against 0.150), gives a decay rate of +0.025 to +0.049 per time unit over doses 0.5 to 2, against -0.001 to +0.023
 221 at the unmatched gated hazard and +0.023 to +0.050 with no gate at all. At matched mean hazard the sister
 222 concordance above independence is +0.051 for the memory gene and -0.000 for the fast-switching control, against
 223 +0.054 and +0.008 unmatched, so the conclusion that the gate cannot manufacture the kin signature does not
 224 rest on the gate also killing less.

225 **Supplementary Note 14: Exact stationary laws for growing, dividing** 226 **populations**

227 The kernels of Supplementary Note 2 are validated against stationary laws for a single cell at fixed volume. That
 228 leaves the layer above them untested against theory: growth, gene replication, partitioning at division, and the
 229 difference between watching one lineage and taking a snapshot of a population. Exact solutions for precisely
 230 this class of model exist. Beentjes, Perez-Carrasco and Grima solve the stationary distribution with bursty
 231 production, DNA replication, binomial partitioning at mitosis and Erlang or general interdivision times, and
 232 give different closed forms for the single-lineage and the population-snapshot setting [9]; Jia and Grima solve the
 233 extended telegraph model with volume-dependent synthesis, dosage compensation, partitioning, interdivision-time
 234 variability and cell-size control, again separately for lineages and populations [10]. The difference between the two
 235 settings is the ergodic principle: a snapshot of a growing population over-weights cells that have recently divided,
 236 and therefore recently lost half their molecules [7,11].

237 Four analytically solvable members of that class were simulated with the population layer and com-
 238 pared with the exact solutions in both modes (`paper/scripts/fig9_validation.jl` for the simulations,
 239 `paper/scripts/exact_solutions.py` for the solutions). The single-lineage mode uses the **MotherMachine**
 240 population control, which keeps one daughter at random at every division, so the recorded cells are statistically
 241 independent lineages; the population mode uses free branching growth with uniform subsampling, which preserves
 242 the snapshot distribution.

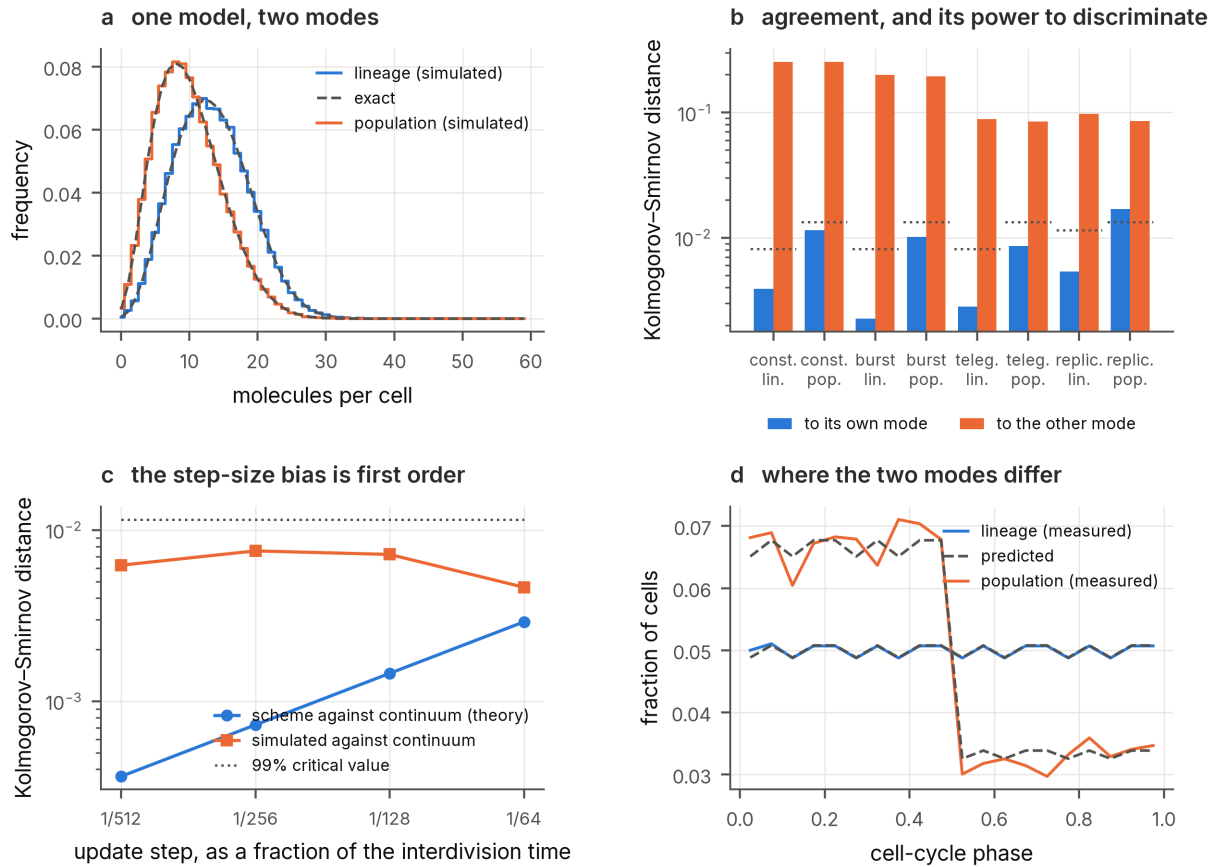
243 **Memoryless interdivision times.** For the first three models the interdivision time is exponential, so the
 244 number of molecules is a Markov process on its own and both stationary laws are available in closed form. Writing
 245 the generating function as $G(u) = \sum_j c_j u^j$ with $u = z - 1$, the factorial moments obey

$$c_j = \frac{\sum_i \beta_i c_{j-i}}{\gamma j + \lambda (1 - 2^{-j})} \quad (\text{lineage}), \quad c_j = \frac{\sum_i \beta_i c_{j-i}}{\gamma j + \lambda (2 - 2^{1-j})} \quad (\text{population}),$$

246 where γ is the decay rate, λ the division rate and β_i the coefficient of u^i contributed by production (for bursts
 247 of fixed size b at rate k_b , $\beta_i = k_b \binom{b}{i}$). For the telegraph model the same expansion gives a two-by-two linear
 248 system per order, because the promoter state is inherited rather than partitioned. The mean is $k/(\gamma + \lambda/2)$ along

a lineage and $k/(\gamma + \lambda)$ in a population, so the two differ by a third at $\gamma = \lambda$ and the comparison has ample power to tell them apart. Because the simulator advances cells by a fixed step dt and divides them at the end of a step, the interdivision time it realises is geometric on that grid; the full pmf of the scheme is therefore the leading eigenvector of $[(1-p)I + \kappa pB]e^{dtA}$ with $p = 1 - e^{-dt/T}$, A the reaction generator, B binomial thinning and $\kappa = 1$ for a lineage or 2 for a population, which is what the simulations are compared against; it agrees with the closed-form moments above as $dt \rightarrow 0$.

Deterministic cycle with gene replication. The fourth model has a deterministic cycle, exponential volume growth, volume-scaled synthesis and gene replication at mid-cycle. Production is linear in volume and copy number and decay is first-order, so the count distribution at every cycle phase is exactly Poisson, with a mean that satisfies a linear recursion over the update steps and is halved at division. The two modes then differ only in how cycle phases are weighted, uniformly along a lineage and towards younger cells in a population; the population weights were obtained by iterating the age map of the scheme from the founder ages actually used, so nothing is assumed about convergence to a stable age distribution. The measured age distribution is compared with that prediction alongside the counts.



263

Supplementary Fig. 6 | The population and lineage layers against exact solutions. **a**, Simulated and exact stationary distributions for constitutive production with a memoryless interdivision time, in single-lineage and population-snapshot mode. **b**, Kolmogorov-Smirnov distance to the exact law of the matching mode, and to the exact law of the other mode, for every model. The dotted line is the 99% critical value for an independent sample of that size, which is exact for the lineage bars and an under-estimate for the population ones, whose cells share ancestors. **c**, Kolmogorov-Smirnov distance against the update step: the theoretical curve is the distance between the stationary law of the scheme and that of the continuous-time model, which is what the discretisation

270

costs, and the measured one sits at the Monte Carlo floor throughout. **d**, Measured and predicted distribution of cell-cycle phase, for the model with gene replication. Along a lineage the phase is uniform. In the population it is the two-level step that a deterministic cycle produces: the snapshot is taken half a cycle after a whole number of generations, and the cells younger than that are the ones whose ancestors have divided once more, so they are twice as numerous. The position of the step is set by when the snapshot is taken and has nothing to do with the replication point, which is also at mid-cycle. The prediction is obtained by iterating the age map of the scheme from the founder ages used.

Along single lineages, where the mother-machine control makes the recorded cells independent and the Kolmogorov-Smirnov test applies as it stands, all 4 samples match the exact law of their mode (distances 0.0023 to 0.0054 against 99% critical values of 0.0081 to 0.0115). A population snapshot is not an independent sample, because its cells share ancestors, so the independent critical value understates the true one and the population mode is judged on the agreement of its replicate means with the exact mean. Those means sit within 2.5 standard errors of the exact values (18 of the 20 individual samples also fall below the independent critical value, the largest distance being 0.0170). All four population means are nonetheless slightly low, by 0.03% to 0.70%; no single deviation is resolved, the largest being 2.5 standard errors ($p = 0.07$, Student's t), but the common sign is not accounted for by the discretisation, the subsampling cap or the burn-in, each of which is excluded in Supplementary Note 14. Scored against the exact law of the other mode, the same samples give distances at least 6 times larger, so the comparison has ample power to tell the two settings apart. The two differ because a snapshot of a growing population over-weights cells that have just divided and so just lost half their molecules: the mean molecule number is constitutive production 13.33 against 10.00; bursts of four molecules 13.33 against 10.00; two-state promoter 8.89 against 6.67; volume-scaled synthesis with gene replication 25.64 against 23.96 along a lineage and in a population respectively.

Discretising the interdivision time to the update step is the only bias in the memoryless-timer comparison, and its size can be computed without simulating anything, as the distance between the stationary law of the scheme and that of the continuous-time model: 0.0004 at $dt = 1/512$, 0.0007 at $dt = 1/256$, 0.0015 at $dt = 1/128$, 0.0029 at $dt = 1/64$. It is first order in the step and already below the 99% critical value at this sample size (0.0115) for every step tested, so a step of a few percent of the interdivision time is enough for the discretisation to be undetectable here. The small common negative offset of the population means is not explained by that discretisation: the exact law each population sample is scored against is the stationary law of the scheme at the simulated step, and solving it at steps from $1/64$ to $1/512$ of the interdivision time moves the exact population mean by less than one part in 10^9 , while it moves the lineage mean by the amounts tabulated above. Nor is it the subsampling cap, which draws cells uniformly without replacement from a snapshot whose division and thinning are independent of the molecule number, so the retained cells are exchangeable; nor the burn-in, since the population mean relaxes at the sum of the decay and growth rates and twelve generations leave a transient far below the observed offset. On the evidence available it is the scatter of five runs on four degrees of freedom, and an offset of this size is neither established nor excluded. In the model with gene replication the measured distribution of cell-cycle phase differs from the predicted one by a total variation distance of 0.001 (lineage) and 0.020 (population) over twenty bins, so the age structure that produces the snapshot weighting emerges from the branching dynamics rather than being imposed on it.

Supplementary Table 5. Simulated against exact stationary laws. Lineage rows are one sample of 40,000 cells; population rows are the mean over 5 independent runs of 15,000 cells each. The quoted error on a population mean is the standard error of that average, the spread between the 5 run means divided by the square root of 5, and on a lineage mean it is the standard error within the single sample; "mean in s.e." is the difference from the

exact mean in those units. “Largest KS” is the worst Kolmogorov-Smirnov distance over the samples of the row,
and the independent critical value beside it is exact for the lineage rows and an under-estimate for the population
rows.

Model	Mode									KS,
		Mean (sim.)	s.e.	Mean (exact)	s.d. (sim.)	s.d. (exact)	Mean in s.e.	Largest KS	Indep. 99%	other mode
constitutive	lineage	13.309	0.027	13.329	5.442	5.432	-0.73	0.0039	0.0081	0.2537
constitutive	population	9.980	0.028	10.000	4.929	4.930	-0.71	0.0115	0.0133	0.2524
bursty	lineage	13.329	0.036	13.329	7.193	7.163	+0.01	0.0023	0.0081	0.1993
bursty	population	9.977	0.033	10.000	6.442	6.437	-0.70	0.0101	0.0133	0.1940
two-state	lineage	8.901	0.051	8.886	10.101	10.110	+0.30	0.0028	0.0081	0.0882
two-state	population	6.620	0.019	6.667	8.419	8.430	-2.45	0.0086	0.0133	0.0845
replication	lineage	25.546	0.057	25.640	8.057	8.073	-1.64	0.0054	0.0115	0.0973
replication	population	23.956	0.077	23.962	7.514	7.485	-0.08	0.0170	0.0133	0.0854

- Iyer A, Alva A, Granada AE, Chakrabarti S. Inheritable cell-states shape drug-persister correlations and population dynamics in cancer cells. *PLoS Comput Biol.* 2025;21: e1013446. doi:10.1371/journal.pcbi.1013446
- Bernton E, Jacob PE, Gerber M, Robert CP. Approximate Bayesian computation with the Wasserstein distance. *J R Stat Soc Series B Stat Methodol.* 2019;81: 235–269. doi:10.1111/rssb.12312
- Emert BL, Cote CJ, Torre EA, Dardani IP, Jiang CL, Jain N, et al. Variability within rare cell states enables multiple paths toward drug resistance. *Nat Biotechnol.* 2021;39: 865–876. doi:10.1038/s41587-021-00837-3
- Shaffer SM, Dunagin MC, Torborg SR, Torre EA, Emert B, Krepler C, et al. Rare cell variability and drug-induced reprogramming as a mode of cancer drug resistance. *Nature.* 2017;546: 431–435. doi:10.1038/nature22794
- Bertaux F, Marguerat S, Shahrezaei V. Division rate, cell size and proteome allocation: Impact on gene expression noise and implications for the dynamics of genetic circuits. *R Soc Open Sci.* 2018;5: 172234. doi:10.1098/rsos.172234
- Piho P, Thomas P. AgentBasedModeling.jl: A tool for stochastic simulation of structured population dynamics. *PLoS Comput Biol.* 2026;22: e1014366. doi:10.1371/journal.pcbi.1014366
- Thomas P, Shahrezaei V. Coordination of gene expression noise with cell size: Analytical results for agent-based models of growing cell populations. *J R Soc Interface.* 2021;18: 20210274. doi:10.1098/rsif.2021.0274
- Lasri A, Shahrezaei V, Sturrock M. Benchmarking imputation methods for network inference using a novel method of synthetic scRNA-seq data generation. *BMC Bioinformatics.* 2022;23: 236. doi:10.1186/s12859-022-04778-9
- Beentjes CHL, Perez-Carrasco R, Grima R. Exact solution of stochastic gene expression models with bursting, cell cycle and replication dynamics. *Phys Rev E.* 2020;101: 032403. doi:10.1103/PhysRevE.101.032403
- Jia C, Grima R. Coupling gene expression dynamics to cell size dynamics and cell cycle events: Exact and approximate solutions of the extended telegraph model. *iScience.* 2023;26: 105746. doi:10.1016/j.isci.2022.105746
- Thomas P. Making sense of snapshot data: Ergodic principle for clonal cell populations. *J R Soc Interface.* 2017;14: 20170467. doi:10.1098/rsif.2017.0467

- 339 12. Piho P, Thomas P. Feedback between stochastic gene networks and population dynamics enables cellular
340 decision-making. *Sci Adv.* 2024;10: eadl4895. doi:10.1126/sciadv.adl4895
- 341 13. Pan X, Li H, Zhang X. TedSim: Temporal dynamics simulation of single-cell RNA sequencing data and
342 cell division history. *Nucleic Acids Res.* 2022;50: 4272–4288. doi:10.1093/nar/gkac235
- 343 14. Jones MG, Khodaverdian A, Quinn JJ, Chan MM, Hussmann JA, Wang R, et al. Inference of single-cell
344 phylogenies from lineage tracing data using Cassiopeia. *Genome Biol.* 2020;21: 92. doi:10.1186/s13059-
020-02000-8
- 345 15. Harmange G, Reyes Hueros RA, Schaff DL, Emert B, Saint-Antoine M, Kim LC, et al. Disrupting cellular
346 memory to overcome drug resistance. *Nat Commun.* 2023;14: 7130. doi:10.1038/s41467-023-41811-8
- 347 16. Oren Y, Tsabar M, Cuoco MS, Amir-Zilberstein L, Cabanos HF, Hütter J-C, et al. Cycling cancer persister
348 cells arise from lineages with distinct programs. *Nature.* 2021;596: 576–582. doi:10.1038/s41586-021-03796-6
- 349 17. Sukys A, Grima R. Cell-cycle dependence of bursty gene expression: Insights from fitting mechanistic
350 models to single-cell RNA-seq data. *Nucleic Acids Res.* 2025;53: gkaf295. doi:10.1093/nar/gkaf295
- 351 18. Zhang Z, Zabaikina I, Nieto C, Vahdat Z, Bokes P, Singh A. Stochastic gene expression in proliferating
352 cells: Differing noise intensity in single-cell and population perspectives. *PLoS Comput Biol.* 2025;21:
e1013014. doi:10.1371/journal.pcbi.1013014
- 353 19. Okochi Y, Sawazaki Y, Kondo Y, Naoki H. Inferring division-associated stochasticity from time-series
354 single-cell transcriptomes. *bioRxiv.* 2026. doi:10.64898/2026.04.14.718485
- 355 20. Pratapa A, Jalihal AP, Law JN, Bharadwaj A, Murali TM. Benchmarking algorithms for gene regulatory
356 network inference from single-cell transcriptomic data. *Nat Methods.* 2020;17: 147–154. doi:10.1038/s41592-
019-0690-6
- 357 21. Dibaeinia P, Sinha S. SERGIO: A single-cell expression simulator guided by gene regulatory networks. *Cell*
358 *Syst.* 2020;11: 252–271. doi:10.1016/j.cels.2020.08.003
- 359 22. Gunnarsson EB, De S, Leder K, Foo J. Understanding the role of phenotypic switching in cancer drug
360 resistance. *J Theor Biol.* 2020;490: 110162. doi:10.1016/j.jtbi.2020.110162
- 361 23. Corigliano M, Di Bernardo A, Cosentino Lagomarsino M, Pompei S. Optimal treatment for drug-induced
362 cancer persisters involves release periods and intermediate drug doses. *PRX Life.* 2025. doi:10.1103/mh5v-
lcyw